Low-Rank Adapters at Scale

Rao, Lindqvist et al. · 2025

Fictional sample paper generated for the ClaimLens demo.

We release the evaluation scripts so that every number can be reproduced. We discuss the limitations of this protocol in the final section. Code and checkpoints are released under an open licence. We discuss the limitations of this protocol in the final section. We keep the tokenizer and context length fixed across every configuration.

We evaluate the final checkpoint rather than the best one seen during training. We report wall-clock time on a single node of eight accelerators. We follow the standard training recipe and report the mean over three seeds. Each configuration is trained for the same number of optimizer steps.

This setting matches the one most often used in practice. Unless noted otherwise, hyperparameters are tuned on a held-out validation split. Variance across seeds is small compared with the differences we report. Prior work reached similar conclusions on smaller models.

Results for individual tasks appear in the appendix. We release the evaluation scripts so that every number can be reproduced. We discuss the limitations of this protocol in the final section. We keep the tokenizer and context length fixed across every configuration.

We report wall-clock time on a single node of eight accelerators. We follow the standard training recipe and report the mean over three seeds. Each configuration is trained for the same number of optimizer steps. Training uses mixed precision with gradient clipping at a norm of one.

Unless noted otherwise, hyperparameters are tuned on a held-out validation split. Variance across seeds is small compared with the differences we report. Prior work reached similar conclusions on smaller models. All runs share the same learning-rate schedule with a short linear warmup.

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We release the evaluation scripts so that every number can be reproduced. We discuss the limitations of this protocol in the final section. We keep the tokenizer and context length fixed across every configuration. The comparison holds the data and the evaluation harness constant.

We follow the standard training recipe and report the mean over three seeds. Each configuration is trained for the same number of optimizer steps. Training uses mixed precision with gradient clipping at a norm of one. Ablations in the appendix vary one factor at a time.

Variance across seeds is small compared with the differences we report. Prior work reached similar conclusions on smaller models. We apply LoRA with rank 16 to every attention and MLP projection. Prior work reached similar conclusions on smaller models. All runs share the same learning-rate schedule with a short linear warmup.

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We discuss the limitations of this protocol in the final section. We keep the tokenizer and context length fixed across every configuration. The comparison holds the data and the evaluation harness constant. This setting matches the one most often used in practice.

Each configuration is trained for the same number of optimizer steps. Training uses mixed precision with gradient clipping at a norm of one. We adapt base models with 7B and 13B parameters. Training uses mixed precision with gradient clipping at a norm of one. Ablations in the appendix vary one factor at a time.

Unless noted otherwise, hyperparameters are tuned on a held-out validation split. Variance across seeds is small compared with the differences we report. Prior work reached similar conclusions on smaller models. All runs share the same learning-rate schedule with a short linear warmup.

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Prior work reached similar conclusions on smaller models. All runs share the same learning-rate schedule with a short linear warmup. Table 2 reports accuracy on our instruction-following suite. All runs share the same learning-rate schedule with a short linear warmup. We evaluate the final checkpoint rather than the best one seen during training.

Table 2

We keep the tokenizer and context length fixed across every configuration. The comparison holds the data and the evaluation harness constant. Accuracy is the fraction of exactly matched answers. The comparison holds the data and the evaluation harness constant. This setting matches the one most often used in practice.

We release the evaluation scripts so that every number can be reproduced. We discuss the limitations of this protocol in the final section. We keep the tokenizer and context length fixed across every configuration. The comparison holds the data and the evaluation harness constant.

We follow the standard training recipe and report the mean over three seeds. Each configuration is trained for the same number of optimizer steps. Training uses mixed precision with gradient clipping at a norm of one. Ablations in the appendix vary one factor at a time.

Variance across seeds is small compared with the differences we report. Prior work reached similar conclusions on smaller models. All runs share the same learning-rate schedule with a short linear warmup. We evaluate the final checkpoint rather than the best one seen during training.

We discuss the limitations of this protocol in the final section. We keep the tokenizer and context length fixed across every configuration. The comparison holds the data and the evaluation harness constant. This setting matches the one most often used in practice.

We keep the tokenizer and context length fixed across every configuration. The comparison holds the data and the evaluation harness constant. On instruction-following suites, LoRA recovers 98–100% of full fine-tuning accuracy across all model sizes tested. The comparison holds the data and the evaluation harness constant. This setting matches the one most often used in practice.

Training uses mixed precision with gradient clipping at a norm of one. Ablations in the appendix vary one factor at a time. LoRA recovers nearly all of full fine-tuning accuracy across our evaluation suite. Ablations in the appendix vary one factor at a time. Results for individual tasks appear in the appendix.

All runs share the same learning-rate schedule with a short linear warmup. We evaluate the final checkpoint rather than the best one seen during training. Across all model sizes, LoRA stays within one point of full fine-tuning. We evaluate the final checkpoint rather than the best one seen during training. We report wall-clock time on a single node of eight accelerators.

We follow the standard training recipe and report the mean over three seeds. Each configuration is trained for the same number of optimizer steps. Training uses mixed precision with gradient clipping at a norm of one. Ablations in the appendix vary one factor at a time.

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Each configuration is trained for the same number of optimizer steps. Training uses mixed precision with gradient clipping at a norm of one. Ablations in the appendix vary one factor at a time. Results for individual tasks appear in the appendix.

Prior work reached similar conclusions on smaller models. All runs share the same learning-rate schedule with a short linear warmup. We evaluate the final checkpoint rather than the best one seen during training. We report wall-clock time on a single node of eight accelerators.

We keep the tokenizer and context length fixed across every configuration. The comparison holds the data and the evaluation harness constant. This setting matches the one most often used in practice. Unless noted otherwise, hyperparameters are tuned on a held-out validation split.

Training uses mixed precision with gradient clipping at a norm of one. Ablations in the appendix vary one factor at a time. Results for individual tasks appear in the appendix. We release the evaluation scripts so that every number can be reproduced.

All runs share the same learning-rate schedule with a short linear warmup. We evaluate the final checkpoint rather than the best one seen during training. We report wall-clock time on a single node of eight accelerators. We follow the standard training recipe and report the mean over three seeds.

The comparison holds the data and the evaluation harness constant. This setting matches the one most often used in practice. Unless noted otherwise, hyperparameters are tuned on a held-out validation split. Variance across seeds is small compared with the differences we report.